

Advanced Deep Learning — Project Proposal

Flow Matching for Image Generation

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Goal and Modelization

The goal of this project is to implement and study *Conditional Flow Matching (CFM)*, a recent generative modeling framework introduced by Lipman et al. [1]. Flow matching learns a continuous vector field $v_\theta(x, t)$ that transports samples from a Gaussian noise distribution ($t=0$) to the data distribution ($t=1$). Following the optimal transport formulation proposed in [1] and further developed in [5], the training procedure defines straight-line interpolations $x_t = (1-t)z_0 + tx_1$ between noise z_0 and data x_1 , and trains a neural network to regress the velocity $u = x_1 - z_0$ at each point along the path. This formulation is closely related to the Rectified Flow framework of Liu et al. [2], which also learns straight-path flows and was later scaled to high-resolution image synthesis in [6]. At inference, new samples are generated by solving the ODE $dx/dt = v_\theta(x, t)$ from $t=0$ to $t=1$ using a standard numerical solver (Euler or Runge-Kutta). The straight-path formulation means fewer integration steps are needed compared to the curved paths in diffusion models [3], making generation significantly faster.

We will implement a U-Net-based flow matching model and systematically compare it against pretrained DDPM [3] and DDIM [4] baselines on image generation quality. Additionally, following the suggestion of our teaching assistant, we will explore *ODE inversion for image editing*: since the flow defines a deterministic ODE, we can invert it (solving from $t=1$ to $t=0$) to recover the noise vector corresponding to a given real image, then manipulate this noise to perform controlled image editing.

Link with Course Concepts

This project connects to several course chapters. **Session 7 (Generative Models)**: flow matching [1] is a direct successor to diffusion models [3] and normalizing flows, both covered in the course; we compare all three frameworks and situate CFM within the generative modeling landscape. **Session 1 (Optimization)**: the training involves a simple regression loss on the velocity field (as formulated in [1, 2]), and generation requires choosing an ODE solver and step count — a tradeoff between speed and accuracy that we study systematically. **Session 4 (Frugal AI)**: the core advantage of flow matching over diffusion models [3, 4] is computational efficiency at inference — achieving comparable quality with 5–10× fewer neural network evaluations, which we will quantify in our experiments.

Dataset

We will use **CIFAR-10** (60,000 32×32 color images, 10 classes) as the primary benchmark, which is the standard dataset for comparing generative models across [1], [2], [3], and [4]. For 2D toy visualizations of the learned vector field, we will use synthetic datasets (moons, circles, spirals) generated with scikit-learn.

Experiment Plan and Baselines

Our experiments are organized in three parts.

(1) Generative quality comparison. We compare our trained CFM model [1] against pretrained baselines: a **DDPM** [3] using publicly available pretrained checkpoints, and **DDIM** [4] sampling at 10, 20, 50, and 100 steps (accelerated sampling applied to the same pretrained DDPM). Our CFM model uses the same U-Net backbone architecture and is trained from scratch on CIFAR-10. The primary metric is FID score (Fréchet Inception Distance) as a function of the number of function evaluations (NFE), directly measuring the efficiency–quality tradeoff that motivates the flow matching framework [1, 2].

(2) ODE inversion and image editing. Since the CFM model defines a deterministic ODE, we can invert it to map a real image x_I back to its corresponding noise vector z_0 by solving the ODE backwards (from $t=1$ to $t=0$). We then perform controlled image editing by perturbing z_0 in various directions and re-generating, following the image manipulation approach enabled by the invertible nature of continuous normalizing flows [1, 2]. We will demonstrate and visualize different types of perturbations and their effects on the generated output.

(3) Visualizations and ablations. On 2D toy data, we visualize the learned vector field and generation trajectories to illustrate the straight-path property of optimal transport CFM [5] vs. the curved paths of diffusion models [3]. We also ablate the ODE solver choice (Euler vs. midpoint vs. RK4) and demonstrate class-conditional generation using class labels injected via cross-attention. Additionally, we visualize latent space interpolations between noise vectors to demonstrate the smooth structure of the learned flow.

References

- [1] Lipman, Y., Chen, R.T.Q., Ben-Hamu, H., Nickel, M., Le, M. *Flow Matching for Generative Modeling*. ICLR, 2023.
- [2] Liu, X., Gong, C., Liu, Q. *Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow*. ICLR, 2023.
- [3] Ho, J., Jain, A., Abbeel, P. *Denoising Diffusion Probabilistic Models*. NeurIPS, 2020.
- [4] Song, J., Meng, C., Ermon, S. *Denoising Diffusion Implicit Models*. ICLR, 2021.
- [5] Tong, A., Malkin, N., Huguët, G., Zhang, Y., et al. *Improving and Generalizing Flow-Based Generative Models with Minibatch Optimal Transport*. TMLR, 2024.
- [6] Esser, P., Kulal, S., Blattmann, A., et al. *Scaling Rectified Flow Transformers for High-Resolution Image Synthesis*. ICML, 2024.